EXATHON INDIA 2026 · MONEY, RISK & REGULATION

# CloseDesk

Governed UPI reconciliation on Exasol Personal. Match three lists in seconds, group what's left by root cause, and let an agent propose fixes that only a human can approve.

github.com/sharma-10-personal/closedesk

MIT licence · synthetic data only

THE PROBLEM

# Every day, three lists disagree — and an analyst untangles them by hand

### NPCI settlement

What the network says happened to each UPI payment.

### Bank ledger

What core banking actually debited or credited.

### Merchant records

What the merchant and gateway think they were paid.

> "I start the day with thousands of mismatches across spreadsheets. By the time I've traced them, the refund deadline for a failed-but-debited customer may already have passed."

THE GOAL

# Start the day with a short list of real exceptions — each with a cause, ₹ at risk and a proposed fix

- Most rows match automatically, inside the database, in seconds.
- Leftovers are classified into named break types, not "unmatched".
- Breaks are grouped by root cause, so one fix can clear thousands of rows.
- The queue is ranked by rupees at risk and deadline, not by row order.
- An agent proposes the fix. A human approves or rejects. Everything is audited.

HOW IT WORKS

# One pipeline, all of it in Exasol SQL

01
### Load
Generate a synthetic day or upload three CSVs. Bulk import with pyexasol.

›

02
### Match
Exact, then amount/time tolerance, then many-to-one batch settlements.

›

03
### Classify
Missing, duplicate, amount mismatch, failed-but-debited, reversal not posted.

›

04
### Group
Root causes by gateway × break type, ranked by ₹ at risk and deadline.

›

05
### Propose
Agent maps each break to an allowed action. Policy gate decides allow, escalate or block.

›

06
### Decide
Human approves or rejects. Decision and audit row written to Exasol.

Nine SQL stages in `sql/`. No Python loops over transactions, no UDFs, no LLM key required.

MEASURED ON A 1,000,000-TRANSACTION DAY (~1M ROWS PER SOURCE)

# Fast enough to run while the answer still matters

**99.1%**
AUTO-MATCHED

**6.65s**
9 SQL STAGES, END TO END

**100%**
PRECISION VS ANSWER KEY

**99.79%**
RECALL VS ANSWER KEY

| STAGE | ROWS | TIME |
|---|---|---|
| Exact match | 977,846 | 3.92s |
| Tolerance match | 70 | 0.57s |
| Many-to-one batch | 2,667 | 0.68s |
| Break detection (4 stages) | 31,417 | 1.35s |

Exasol Personal 2026.2.0 · local VM, 2 vCPU · MacBook

FROM ROWS TO CAUSES

# 31,417 breaks. 31 root causes.

| 989,666 | 980,583 | 31,417 | 31 |
| --- | --- | --- | --- |
| NPCI ROWS | MATCHED | BREAKS | ROOT CAUSES |

Each cause is a gateway × break type group, sized by rupees at risk. The analyst fixes the cause once — a gateway clock, a duplicate posting job — instead of chasing thousands of rows.

GOVERNED AGENT

# The agent proposes. A human decides. Exasol remembers.

**Allow · ≤ ₹50,000**

Offered for one-click approval by the analyst.

**Escalate · up to ₹5,00,000**

Flagged for a supervisor before anyone acts.

**Block · above that**

Approve is disabled in the UI and refused in code. Manual process only.

- The agent can only choose from an allow-list of five actions in `config/policy.yaml`.
- Every proposal, approval and rejection is written to `audit_log` with before and after state.
- Explanations come from templates and SQL labels, so it runs with no LLM and no API key.

THE PRODUCT

# Five screens, all reading live Exasol state

### Pipeline

Generate a synthetic day or upload your three CSVs, validated before load. Watch each SQL stage run.

### Investigate

One NPCI reference across all three sources, its match or break, and the proposed fix.

### Inject

Plant a new break into the live data and watch CloseDesk detect and propose a fix in seconds.

### Exceptions

The proposal queue ranked by ₹ at risk, filtered by type, gate and status. Approve or reject.

### Analytics

Funnel, break types, root causes, accuracy against the answer key, and the full audit trail.

### Try it

```
make setup
make full
make demo
```

WHAT IT DOES NOT DO YET

# Honest limits and next steps

## Limits today

83 timing-skew cases that cross midnight are missed (recall 99.79%).

Approval records the decision; it does not post to a core-banking system.

The agent runs as the database owner, not a read-only user.

All data is synthetic. The compensation rate is a placeholder.

## Next

Cross-midnight tolerance window.

Read-only database role for the agent.

Undo for decisions, and posting via a maker-checker integration.

Generate and match 10M+ rows entirely in Exasol SQL.

CLOSEDESK

# Close the day before lunch, not after midnight.

Open source under MIT. Exasol Personal, Python, Streamlit. All transactions synthetic.

github.com/sharma-10-personal/closedesk